Progressive Feature Interleaved Fusion Network for Remote-Sensing Image Salient Object Detection

Pengfei Han, Bin Zhao, and Xuelong Li, *Fellow, IEEE*

***Abstract*— Salient object detection (SOD) has made significant strides in natural scene images (NSIs) in the span of the past few decades. However, extending these approaches for remote-sensing images (RSIs) faces challenges due to their complex backgrounds, complicated edges, irregular topology, and multiscale object variations, which hinder performance. Existing RSI-SOD techniques are unable to accurately detect salient objects while preserving detailed boundaries, and their computational inefficiency limits their practicality. To overcome these challenges, we entail the development of a progressive feature interleaved framework (PROFILE) in RSI-SOD. In particular, we leverage the interleaved association of the convolutional neural network (CNN) and Transformer (IACTer) to obtain global semantic relations and spatial details. To handle object scale variation, we design a lightweight *plug-and-play* multiscale hierarchical channel-spatial collaborative feature enhancement module (MHCCF), which can boost the representation of features regarding the relevant region, while identifying the precise location details about the salient region. Finally, a bi-directional consistency constraint module (BCCM) is developed, which can be integrated into the training of arbitrary SOD and segmentation networks to efficiently locate salient regions with refined structures and clear demarcations. Experiments demonstrate that our PROFILE surpasses 20 cutting-edge SOD methods, proving its ability to enhance the accuracy and integrity of SOD in complex backgrounds, such as illumination and shadows.**

***Index Terms*— Bi-directional consistency constraint, feature enhancement, optical remote-sensing image (RSI), saliency object detection.**

I. Introduction

SALIENT object detection [1] purposes by locating visually distinctive and interesting regions/objects within the image. It has proven to be an effective preprocessing step for a variety of computer vision applications, for instance, image enhancement [2], image match [3], change detection [4], and object detection [5], [6], among others. However, effectively and efficiently utilizing remote-sensing image (RSI) information to achieve accurate salient object detection (SOD) remains a significant challenge.

Manuscript received 8 June 2023; revised 17 October 2023 and 2 December 2023; accepted 3 December 2023. Date of publication 5 December 2023; date of current version 13 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62106183 and Grant 62376222. *(Corresponding authors: Bin Zhao; Xuelong Li.)*

Pengfei Han is with the School of Cybersecurity and the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, Shaanxi 710072, China.

Bin Zhao and Xuelong Li are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, Shaanxi 710072, China (e-mail: bin@nwpu.edu.cn; li@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3339970

Early SOD techniques relied mainly on hand-designed features, like color contrast [7], boundary background [8], and center prior [9] to detect salient objects in images. Nevertheless, such features have yielded insufficient semantic feature information that is required for inferring salient objects, and, therefore, fail to generate satisfactory saliency maps. To address the above issues, several learning-based methods can be developed to capture global contextual features, improving SOD performance by integrating feature mapping on multiple layers of the convolutional neural network (CNN). Although methods such as LVNet [10], SARNet [11], and DAFNet [12] have achieved better results, in complex scenes of cluttered backgrounds and diverse shapes in RSI-SOD, the predicted objects often suffer from inaccurate localization, blurred edges, and incomplete structures.

A. Motivation

To handle the above problem, some researchers have presented methods with boundary feature-based learning or introduced additional edge prediction branches [13], [14], ensuring that the predicted saliency map has finer and clearer boundaries. Due to the intricate interlacing of salient and nonsalient regions, the acquisition of both global and local representations holds significant importance in the challenging realm of RSI-SOD tasks. Several studies [15], [16], [17], [18] have delved into hybrid CNN–Transformer architectures, blending efficient convolutional blocks with potent Transformer blocks. Most of these hybrids replace shallow Transformer blocks with convolutional ones, aiming to mitigate the high computational cost of early-stage self-attention. However, these simplistic stacking strategies impede their ability to strike a better balance between accuracy and efficiency. Additionally, SOD facilitates the processing of image compression to achieve retention of vital information and filtering of useless information, improving the efficiency of data transmission and reducing the amount of transmitted data. However, few investigations have been conducted on SOD in optical RSIs, which is of utmost importance for RSI compression.

Despite substantial progress in natural scene image (NSI)-SOD, the RSI-SOD still confronts various obstacles that severely limit detection capabilities [19], which can be depicted as follows.

1) Salient objects in RSIs are frequently contaminated by intricate backgrounds and imaging circumstances (i.e., shooting duration and illumination intensities) that may cause noise nuisance (e.g., shadows and intense exposures). This, in turn, amplifies the challenge of

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

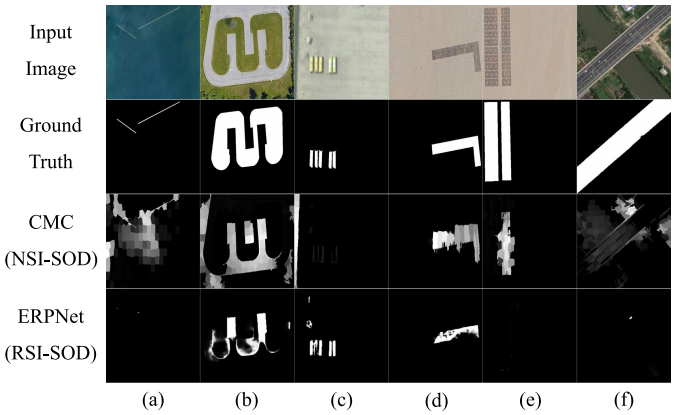


Fig. 1. SOD performance in challenging remote-sensing scenes. (a)–(f) Samples of remote sensing saliency detection in different scenarios. The CMC is the NSI-SOD method, and the ERPNet is the RSI-SOD method.

precisely pinpointing the salient objects and their corresponding regions in RSIs. For example, CMC [20] and ERPNet [21] fail to locate the salient region in Fig. 1 of the left three columns.

2) Since RSIs are often taken from various ground sampling distances with respect to diverse capture devices, the salient objects/regions in optical RSIs exhibit intricacies in their structures and topologies that surpass those present in NSIs. Consequently, this poses new hurdles to accurately apprehending complete regions of salient objects and upholding sharp boundaries. For example, missing or incomplete results of SOD among the last three columns in the Fig. 1. Particularly, ERPNet, specifically designed for RSI-SOD cannot even localize to salience objects in Fig. 1(f). Confronting the aforementioned opportunities and challenges, this article aims to explore a new perspective on integrating Transformer and convolutional blocks tailored for RSI-SOD.

After a comprehensive analysis of the research presented, we explore an instantaneous and practical progressive feature interleaved framework (PROFILE) approach. It aims to precisely identify salient regions while maintaining their intricate boundaries. Specifically, to efficiently capture both accurate global semantic relations and spatial details, we deploy an Interleaved Association of CNN and Transformer (IACTer). IACTer allows the CNN to inherit global information and compensate for their common deficiencies through the interleaved association of CNN and Transformer. Moreover, to enhance the learning of the representation of foreground objects in RSIs affected by background interference, we introduce a lightweight *plug-and-play* multiscale hierarchical channel-spatial collaborative feature enhancement module (MHCCF). By gradually integrating multilevel image features into the network, the MHCCF improves the network's modeling capability for spatial and channel information. The result is a notable improvement in the feature representation of relevant areas, which addresses the issue of object scale variation in optical RSIs. Finally, to tackle the issue of incomplete prediction results with coarse and ambiguous boundary details, we devise the bi-directional consistency constraint module (BCCM), which merges pixel-level bi-directional consistency with patch-level structure bi-directional consistency to enhance the effectiveness of the proposed model. The BCCM can be incorporated into any SOD or segmentation model to obtain complete salient objects with well-defined boundaries.

B. Contributions

From the foregoing, the merits ascribed to this article the following points can be summarized.

1) The IACTer employs a novel interleaved association of the CNN and Transformer, enabling the effective capture of spatial details and semantic relations. This architecture ensures accurate prediction of saliency in high-resolution RSIs without compromising real-time efficiency.
2) The MHCCF progressively integrates multilevel image features to enhance feature representation of relevant areas and address object scale variation in optical RSIs.
3) The BCCM provides implicit supervision over the region of interest and its boundaries and can be seamlessly integrated into any SOD and image segmentation network.
4) The suggested approach achieves optimal detection performance in terms of both quantity and quality compared to 20 cutting-edge methodologies, obviating the necessity for any antecedent preprocessing methodologies or subsequent processing techniques.

C. Organization

The following is the order of the article's succeeding sections. Section II offers a succinct, yet methodical appraisal of the various SOD techniques. Section III expounds upon the PROFILE method, outlining the details comprehensively. Furthermore, Section IV showcases the assessment of current cutting-edge methods via various metrics, culminating with a summary in Section V.

II. Related Work

During the past several decades, an array of theories has been employed in the advancement of saliency models. This section briefly reviews the relevant literature on the RSI-SOD in both traditional methods and deep-learning-based methods.

A. Traditional SOD Methods

Traditional SOD techniques relied mainly on hand-designed features to identify visually appealing objects or regions in images, which typically extract low-level features like color contrast [7], [22], boundary background [8], [23], and center prior [9], [24]. For instance, Liu et al. [25] incorporated multiscale contrast, center-surround histograms, and spatial color distribution into a conditional random fields model. While substantial inroads have been advanced in SOD for NSIs, the RSI-SOD is still in its infancy, with only a limited number of studies available. For instance, Zhao et al. [26] introduced a saliency model that utilizes sparsity-guided techniques to generate saliency maps by incorporating semantic and contextual information. Zhu et al. [27] integrated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

a differential threshold-based psychological feature into a supervised Markov random field model. Peng et al. [28] proposed a structural matrix decomposition-based SOD technique that uses sparse and Laplacian regularization to measure the saliency difference between an object and a background region. Zhang and Yang [29] suggested a QFT-based frequency-domain methodology to efficiently and expeditiously identify significant regions in RSIs. Additionally, Ma et al. [30] implemented ROI detection by analyzing super pixel-to-pixel saliency. Zhang et al. [31] suggested a novel saliency analysis model that analyzes global and local saliency to extract saliency areas in RSIs. However, these hand-crafted methods yield limited capabilities in capturing high-level semantic information, which is crucial for SOD tasks involving complex backgrounds and rich semantics. Yet, substantially decreased research focus on these methods can be attributed to the emergence of deep learning.

B. Deep-Learning-Based SOD Methods

Recent studies [18], [32] have made significant progress in RSI-SOD, despite the formidable challenges it presents compared to natural imagery. These challenges include diverse meteorological conditions, complex backgrounds, varied orientations, and scale variations. As a result, researchers have constructed RSI-SOD datasets to cope with these issues. For example, the earliest available public RSI-SOD database set, known as ORSSD, and its extension, EORSSD, were created in [10] and [12]. Furthermore, in [12], Zhang et al. proposed an attentional fluid network that can learn more discriminative saliency characteristics with a well-designed global context-aware attention module. Liu et al. [33] used saliency bounding boxes for weakly supervised salient object identification. Such approaches are based on hand-designed features for producing fake salience labels, which are devoid of semantic information. To eliminate these demerits, a two-stage SOD architecture was introduced by efficiently mining saliency cues in [34]. Despite these advancements, SOD models frequently bypass the capability of accurately detecting salient objects owing to the omission of edge information from complicated and disorganized backdrops.

To address this issue, Zhou et al. [21] presented an edge-assisted recursive model that can accurately identify salient objects in optical RSIs. Yasarla et al. [35] introduced a theoretically basic self-supervised SOD approach. Wang et al. [36] exploited the traditional hand-crafted SOD process with noise labels to potentiate unsupervised multisource uncertainty mining. Label values may lead to erroneous predictions, but pseudo-labels are unanimous. To address this issue, a novel approach dubbed learning salient feature (LSF) accomplishes unsupervised SOD via LSF from the data itself in [37]. Qin et al. [38] suggested an edge-aware framework that accurately detects salient regions, but the high complexity of parameter and model requirements slows inference speed. Li et al. [39] created a shallow and narrow network SOD model for efficiency and accuracy. Additionally, Li et al. [40] devised a dual-branch feature fusion object detection framework that can realize the positioning and recognition of target spatial location even in demanding settings, such as rotational changes and blurred appearance. Furthermore, Zhou et al. [41] presented an edge-aware network to improve SOD detection performance, which implements multiscale features by aggregating image pyramids, feature pyramids, and edge information. While these approaches have substantially enhanced the benchmarks for detecting significant objects, there remains a noteworthy scope for refinement with regard to the precision of fine-grained segmentation and boundary recovery accuracy in complex scenarios.

III. Proposed Method

In this section, we provide a comprehensive elucidation of the proposed PROFILE model, as depicted in Fig. 2, which comprises the IACTer, MHCCF, and BCCM modules. The IACTer module strategically integrates two distinct network architectures to extract local multiscale features and global multiscale features from the input image. This strategic approach enriches group convolution with a plethora of multiscale features. Furthermore, we have introduced the MHCCF module, meticulously crafted to fuse multitiered local features from both the channel and spatial dimensions. This careful preparation enables a distinctive comparison with multiscale global features, culminating in the derivation of the final differential map. To ensure precise fine-tuning of the edges within the differential map, we have conceived the BCCM module, which can effectively mitigate the influence of upsampling and downsampling on edge features. Consequently, it imposes additional constraints on the differential map, bringing its edge features into closer alignment with the ground truth.

A. Preliminary

Owing to the induction bias, CNN-based models can converge quickly but fail to have the capability of capturing long-range dependencies, specifically with optical RSIs. With more flexible unstructured data modeling capabilities, the Transformer can adaptively extract local/global contextual relations [42], [43]. However, Transformers have intrinsic downsides. In the training phase, they entail a secondary complexity of computation that is relevant to the image resolution or the number of markers, a huge attentional computation when dealing with long sequences of tokens, slow convergence, and failure to perform well on small datasets. Additionally, as evidenced in [44], the image can be mathematically stated as follows:

$$I = I_H + I_L \tag{1}$$

where I represents the input RSI. I_H denotes a high-frequency component, which is defined as unreliable information, that is, texture and detail. I_L expresses the low-frequency component, which is specified as reliable information, encompassing the global structure and contours.

Motivated by the analysis above, the proposed method follows the "divide and conquer" motif as its core idea for designing a progressive feature interleaved fusion network. This network effectively harnesses the supplementary

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

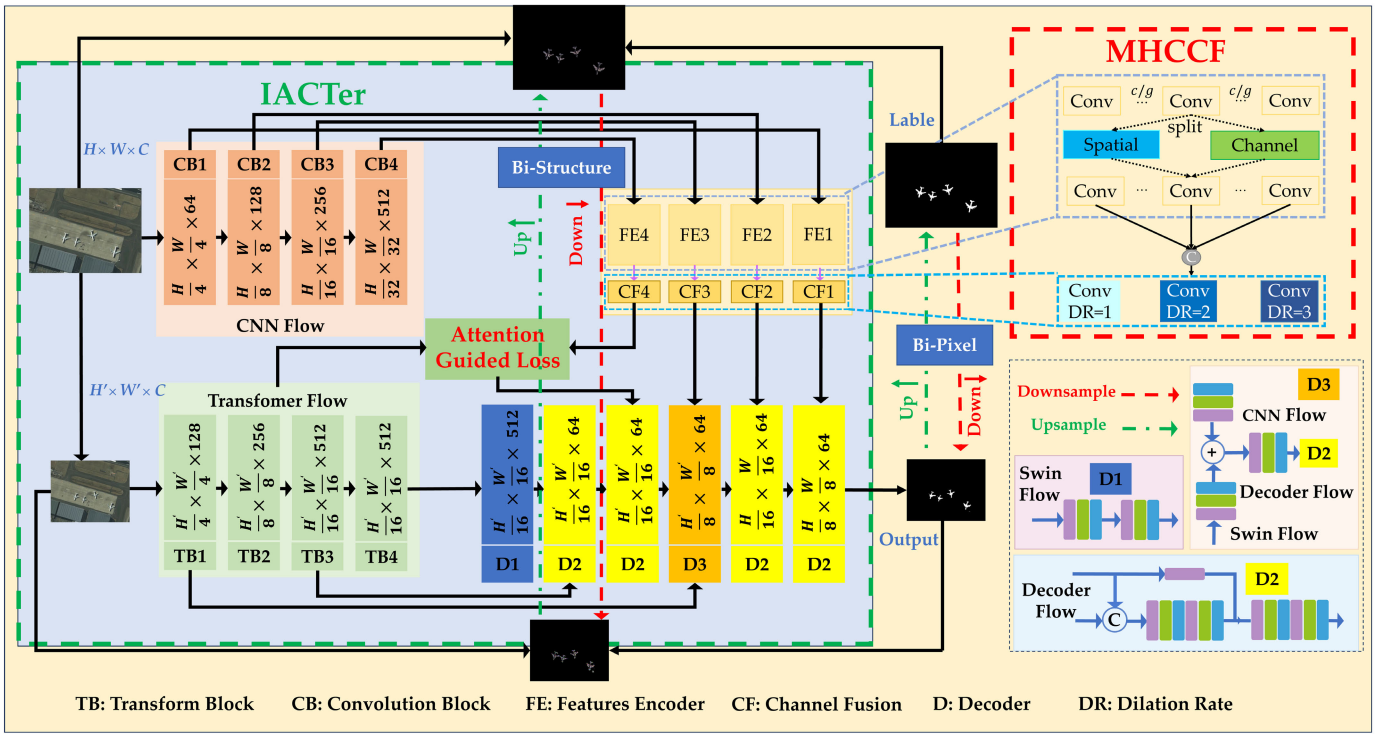


Fig. 2. Flowchart illustration about the proposed PROFILE for RSI-SOD.

characteristics of both CNNs and Transformers. Additionally, we have discovered that it inclines to be the point in the input embeddings from the high-level feature map which identifies a small instance, while it confers a tendency to be the point from the low-level feature map that recognizes a large instance.

B. Interleaved Association of the CNN and Transformer (IACTer)

To leverage this observation, we have designed a network consisting of two encoding streams: the image detail encoder and the semantic information encoder. By separating the encoding streams in this way, our network can acquire both the subtle details of an image and the broader context in which it exists. This approach allows for a more nuanced and accurate understanding of the image. Since RSIs are captured at different ground sampling distances, the scale of objects can vary dramatically with different image resolutions, hindering the efficacy of SOD. To tackle the aforesaid matter, the Inception v3 architecture has been selected as the image detail encoder, as it can extract more diverse and detailed local texture features across multiple scales compared to ResNet-18 [45]. The CNN flow of the system can then attain multiscale hierarchical detail features by analyzing the output of various layers within the network. By providing input images $I_L \in \mathbb{I}^{3\times448\times448}$, we can derive four distinct levels of local features: $\mathbf{I}_1^{64\times112\times112}$, $\mathbf{I}_2^{128\times56\times56}$, $\mathbf{I}_3^{256\times28\times28}$, and $\mathbf{I}_4^{512\times14\times14}$. These features are comprised of copious high-frequency information of the image, due to the Inception network's unique multiscale convolutional kernel. To merge the high-frequency information across channels, we further compress the image channel features of each output branch. This results in obtaining $\mathbf{I}_1^{'64\times512\times512}$, $\mathbf{I}_2^{'128\times256\times256}$, $\mathbf{I}_3^{'256\times128\times128}$, and $\mathbf{I}_4^{'512\times64\times64}$, respectively, which can guide the global semantic relations within the image obtained through the Transformer flow.

For the semantic information encoder, the Swim Transformers are introduced, which possess a superior capability of capturing low-frequency features with the possibility of supplying global information (i.e., shape and structure), and establishing the connection between image blocks. Similar to the CNN flow, the Transformer flow receives input images $I_H \in \mathbb{R}^{3\times224\times224}$ and produces four stages of output, denoted as $\mathbf{S}_1^{128\times56\times56}$, $\mathbf{S}_2^{256\times28\times28}$, $\mathbf{S}_3^{512\times14\times14}$, and $\mathbf{S}_4^{512\times14\times14}$, respectively. These four stages of features capture rich high-frequency information. Moreover, the dual-encoders adeptly capture both the global semantic relations and the intricate spatial detail information within the image during the encoding phase, mutually guiding and complementing each other to better identify salient objects.

To better integrate the information of the Transformer flow with that of the CNN flow, the widely deployed attention-guided loss is employed to remedy errors common to both the Transformer and CNN flows, as suggested by [17] and [18]. The attention-guided loss can be referred to as supplementary materials in [17] and [18]. It allows CNNs to inherit global information and compensate for their common shortcomings through the interleaved association of the CNN and Transformer. Fig. 2 depicts the feature map flow in the decoder block of the proposed architecture. In a staggered arrangement, the CNN flow comes first, then the attention-guided module, and then the decoder module gets the flow of features from the Transformer flow. The decoder block is broken up into sections made up of the D1, D2, and D3 subblocks. The network achieves deeper sampling depth while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

minimizing computational costs, enabling it to adeptly tackle the challenges posed by high-resolution optical RSIs.

C. Multiscale Hierarchical Channel–Spatial Collaboration Feature Enhancement Module

Distinct from PGNet [17], we employ the Inception v3 as the image detail encoder. It enables the acquisition of more intricate and diverse multiscale local texture features of the image, as compared to ResNet, owing to its unique design of multiscale convolution. By analyzing the output of various network layers, we can obtain multiscale hierarchical detail features of the image. However, the features obtained by the original encoder network are relatively coarse-grained and lack adequate fineness of local information in both spatial and channels. Some works [46], [47], [48], [49] fundamentally utilize multiscale feature fusion and multiscale convolution operations to obtain the multiscale features of the object. However, the focus of these works on multiscale features is confined to a singular object structure, lacking consideration for different structural features of the object. Additionally, in this manner, the features obtained may lack the focus on crucial and salient information, which is detrimental to the ultimate precision of detection. Therefore, we present an MHCCF, depicted in Fig. 2, which effectively amplifies and underscores the intricate details of the image.

The utilization of feature enhancement has gained widespread adoption in computer vision. To further augment the multiscale hierarchical image features extracted by the Inception v3 network and to significantly enhance image detail information, we have introduced a multiscale hierarchical feature enhancement module. To elaborate, given a feature map denoted as $X \in \mathbb{R}^{C\times H\times W}$, we partition it into G groups based on the dimension of the channel such that $X = [X_1, \ldots, X_G]$, where $X_k \in \mathbb{R}^{C/G\times H\times W}$. Every subgroup can learn certain semantic features, and an attention module produces the relevant importance coefficients for each subfeature. Typically, each subgroup is further divided into two branches. One branch leverages the inter-relationships of channels to generate channel attention maps, while the other exploits features' spatial relationships to generate spatial attention maps. In addition to optimizing encoder features in the spatial dimension, it also facilitates the enhancement of the channel dimensions of the encoders. Specifically, it is formulated as

$$S_c = \mathbf{P}_{\text{gap}}(f_c) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} f_c(i,j) \tag{2}$$

where $\mathbf{P}_{\text{gap}}$ denotes the global average pooling operation, f_c denotes channel feature representation branch feature maps, and S_c denotes generated channel-wise statistics. Subsequently, we can obtain the output in the following manner:

$$f'_{c-\text{att}} = \sigma(\mathbf{L}_c(S_c))\cdot f_c = \sigma(W_c S_c + b_c)\cdot f_c \tag{3}$$

where W_c and b_c denote the weights and biases of linear layers on each channel, respectively. On the other hand, it allows for highlighting detailed information in the spatial feature. As described below

$$f'_{s-\text{att}} = \sigma(\mathbf{C}_s(GN(f_s)))\cdot f_s = \sigma(W_s GN(f_s) + b_s)\cdot f_s \tag{4}$$

where $\mathbf{C}_s$ denotes the convolution operation and W_s and b_s are parameters with shape $\mathbf{R}^{(C/2G\times H\times W)}$, respectively. When two feature maps have been yielded by dual branches, the ultimate output can be acquired by the subsequent process

$$f_{\text{att}} = f'_{c-\text{att}} \cup f'_{s-\text{att}} \tag{5}$$

where f_{att} is concatenated by the two branches $f'_{c-\text{att}}$ and $f'_{s-\text{att}}$ as input.

To accomplish the ultimate multiscale hierarchical feature enhancement, we have devised a channel fusion module that enables interactions between the features of the groups within G. Each group within G possesses its own distinct spatial and channel attention features. The channel fusion module executes a shuffle operation on the combined features at the channel level, subsequently followed by a multiscale convolution that further fuses the shuffled features. This process facilitates the extraction of more delicate local details, which can complement the semantic features obtained by the Swin Transformer encoder more effectively. The channel fusion operation can be decomposed into the following stages:

$$f'_{\text{att}} = \text{Shuffle}(f_{\text{att}}) \tag{6}$$

$$f = \text{DilatedConv}\left(f'_{\text{att}}\right) = \sum_{i=1,2,3} W_i f'_{\text{att}} + b_i \tag{7}$$

where Shuffle and DilatedConv denote the shuffle operation and the dilated convolution, respectively.

D. Bi-Directional Consistency Constraint Module

Most SOD methods for detecting the most salient object rely on binary cross-entropy (BCE) for their optimization term. However, BCE-optimized models often display inadequate confidence in distinguishing boundary pixels, resulting in saliency maps that lack fine structure or boundary definition. Intersection over union (IoU) [38] was proposed for biased training sets, but it is not designed explicitly for capturing fine structures. Building on this motivation, we propose a BCCM to obtain high-confidence saliency maps and clear boundaries, as shown in Fig. 3. It combines pixel bi-directional consistency and structure bi-directional consistency, both of which are promising to enable implicit optimization of the model at the pixel and patch levels. In fact, the amalgamation of these two terms captures the tradeoff necessitated for balancing the impact of foreground and background pixels, while boosting the detailed contour portrayal of the objects.

1) Bi-Directional Consistency of Structure: For a given image, it is fed to the network for processing to obtain a set of prediction boundaries. These boundaries are then constrained at the patch level with the ground truth through upsampling, while the ground truth is also constrained with the network prediction boundary through downsampling, therefore the bi-directional consistency of the generated results and the reference map has been implemented. Overall, by employing a structure of bi-directional consistency loss, we boost the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

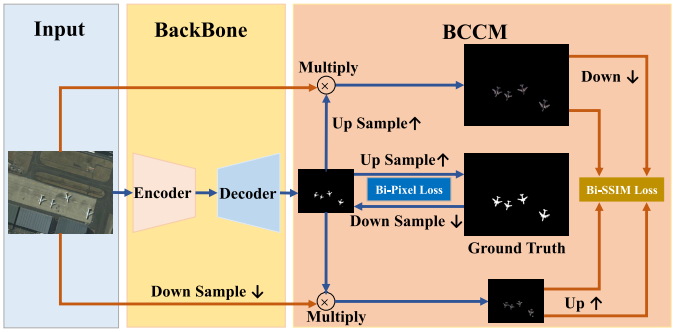


Fig. 3. Detailed structures of BCCM.

detailed contour portrayal of the objects by balancing the impact of foreground and background pixels. Although this approach may result in a somewhat blurry output, it guides the network toward roughly predicting texture and low-frequency details. It can be described as

$$\ell_s = \mathrm{SSIM}\left(\left|m(i,j)\uparrow - \hat{m}(i,j)\right| + \left|m(i,j) - \hat{m}(i,j)\downarrow\right|\right) \quad (8)$$

where ℓ_s denotes structural similarity loss function and $m(i,j)$ represents masked image predict by network. $\hat{m}(i,j)$ denotes the reference masked image. $\uparrow$ and $\downarrow$ indicate upsampling and downsampling using the interpolation method.

2) Bi-Directional Consistency of Pixel: It is likely that the saliency map predicted from the higher stage will have a relatively small spatial dimension, which could exacerbate the problem of stage-scale inconsistency, particularly with regard to the boundary areas of salient objects. To address this issue, we focus on "Do what you can with what you have where you are." Building upon the BCCM, we implicitly model salient edges to enhance predicted saliency maps with clear and accurate boundary details. Additionally, predicted mask images are generated by combining the predicted network boundary with input RSIs. In the same way, a masked image can be created by combining the ground-truth mask label with input RSIs. The description of the bi-directional consistency structure is partitioned into the subsequent definitions

$$\ell_p = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left|x(i,j)\uparrow - \hat{x}(i,j)\right| + \left|x(i,j) - \hat{x}(i,j)\downarrow\right| \quad (9)$$

where $x(i,j)$ signifies the predicted boundary by the network and $\hat{x}(i,j)$ indicates the reference image boundary. The symbols $\uparrow$ and $\downarrow$ signify upsampling and downsampling using the interpolation method, respectively.

E. Optimizing Strategy

IoU is a frequently employed loss function in classification tasks, as widely acknowledged. In SOD, the predominant method involves utilizing the IoU loss to compare the generated saliency results with the reference map at the image level. The loss function's definition is provided below

$$\ell_{\mathrm{IoU}} = 1 - \frac{\sum_{i=1}^{H} \sum_{j=1}^{W} m(i,j)\hat{m}(i,j)}{\sum_{i=1}^{H} \sum_{j=1}^{W} \left[m(i,j) + \hat{m}(i,j) - m(i,j)\hat{m}(i,j)\right]} \quad (10)$$

where $\hat{m}(i,j) \in \{0,1\}$ represents pixel (i,j)'s ground truth, and $m(i,j)$ signifies the salient object of a prediction probability. The input image has dimensions W and H, which represent its width and length, respectively. Furthermore, to gauge the discrepancy that exists between the reference binarized image and the generated results, we have optimized the distortion loss by integrating the mean squared error (MSE), which can be described as

$$\ell_{\mathrm{MSE}} = \left\|m(i,j) - \hat{m}(i,j)\right\|^2. \quad (11)$$

Additionally, the BCE loss has been exploited for error appraisal of the generated saliency results and ground truths, which are specified as

$$\ell_{\mathrm{BCE}} = -\sum_{(i,j)} \left[\hat{m}(i,j)\log(m(i,j)) + \left(1 - \hat{m}(i,j)\right)\log(1 - m(i,j))\right]. \quad (12)$$

Furthermore, we also apply the widely used attention-guided loss to remedy the errors common to both the Transformer and CNN flows. To delve into a detailed explanation of the Attention-Guided Loss ℓ_{ag}, we earnestly encourage readers to refer to the original work by [17] and [18]. At the training stage, we use a combination of three loss functions with a proposed BCCM, and they are optimized jointly via back-propagation using an SGD optimizer. Taking into account the reconstruction and consistency terms, we have constructed the final expression for the loss from three perspectives: pixel level, patch level, and image level. The final expression of the loss is given by

$$\ell_{\mathrm{total}} = \lambda_1\ell_p + \lambda_2\ell_s + \lambda_3\ell_{\mathrm{MSE}} + \lambda_4\ell_{\mathrm{ag}} + \lambda_5\ell_{\mathrm{BCE}} + \lambda_6\ell_{\mathrm{IoU}} \quad (13)$$

where the ultimate optimization factors are balanced among the hyperparameter in terms of λ_1, λ_2, λ_3, λ_4, λ_5, and λ_6. ℓ_p represents the bi-directional consistency of pixel, which can be defined by (9). ℓ_s denotes the bi-directional consistency of structure, which is defined by (8). All hyperparameters are dynamically adjusted throughout the training process via the GradNorm [50], which automatically balances the training of the proposed PROFILE model by adapting gradient magnitudes based on the characteristics of the multitask loss function. For a detailed explanation, we kindly refer readers to the original work authored by [50].

IV. Experiments Results

A. Datasets

To assess and analyze the effectiveness of our proposed algorithm, we perform experiments on two publicly available remote-sensing datasets: ORSSD [10] and EORSSD [12]. Below are detailed descriptions of the two datasets as follows.

1) ORSSD Dataset: The ORSSD dataset is specially designed for RSI-SOD, with 800 images and eight object categories, for example, ship, car, river, pond, bridge, stadium, and beach. Practically, 600 images have been randomly sourced during training, while the remaining 200 are reserved for testing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
AVERAGE ACCURACY COMPARISON ON THE EORSSD DATASET. THE FIRST THREE BEST PREDICTIONS HAVE BEEN RENDERED IN **RED**, **BLUE**, AND **BLACK**

Method	Publication	EORSSD Dataset								
		$MAE \downarrow$	$F_{max} \uparrow$	$F_{mean} \uparrow$	$F_{adp} \uparrow$	$E_{max} \uparrow$	$E_{mean} \uparrow$	$E_{adp} \uparrow$	$S_m \uparrow$	$F_{bw} \uparrow$
DSS [56]	CVPR2017	0.0186	0.7080	0.6010	0.4754	0.9205	0.7647	0.6947	0.7879	0.5721
R3Net [60]	IJCAI2018	0.0171	0.7748	0.6532	0.4309	0.9499	0.8320	0.6476	0.8198	0.5500
RADF [59]	AAAI2018	0.0168	0.7704	0.6822	0.5106	0.9162	0.8597	0.7279	0.8195	0.6116
PoolNet [57]	CVPR2019	0.0206	0.7564	0.6468	0.4770	0.9173	0.8055	0.6852	0.8219	0.5673
EGNet [13]	ICCV2019	0.0109	0.7993	0.7065	0.5545	0.9410	0.8631	0.7566	0.8605	0.6622
LVNet [10]	TGRS2019	0.0145	0.8065	0.7591	0.6501	0.9279	0.8828	0.8465	0.8650	0.7316
GateNet [58]	ECCV2020	0.0095	0.8831	0.8489	0.7329	0.9610	0.9387	0.8743	0.9109	0.8347
GCPA [61]	AAAI2020	0.0103	0.8544	0.8080	0.6931	0.9473	0.9096	0.8647	0.8861	0.7995
ITSD [62]	CVPR2020	0.0106	0.8785	0.8477	0.7651	0.9555	0.9406	0.8986	0.9045	0.8299
U2Net [63]	PR2020	0.0076	**0.8998**	0.8593	0.7444	0.9646	0.9372	0.8989	0.9208	0.8503
DAFNet [12]	TIP2021	**0.0060**	0.8878	0.8093	0.6622	**0.9859**	0.9294	0.8143	0.9176	0.7804
PAKRN [64]	AAAI2021	0.0104	0.8906	**0.8621**	**0.8240**	0.9616	0.9537	0.9250	0.9186	0.8501
SUCA [65]	TMM2021	0.0097	0.8454	0.8168	0.7484	0.9507	0.9251	0.8798	0.8985	0.8092
PGNet [17]	CVPR2022	0.0153	0.7887	0.7775	0.7402	0.9315	0.9178	0.8858	0.8757	———
EMFI [41]	TGRS2022	0.0084	0.8990	**0.8751**	0.8231	0.9706	**0.9601**	**0.9501**	**0.9299**	**0.8689**
MJRBM [55]	TGRS2022	0.0102	0.8907	0.8488	0.7272	0.9635	0.9345	0.8886	0.9191	0.8346
RRNet [54]	TGRS2022	0.0078	**0.9047**	0.8505	0.7189	**0.9715**	0.9330	0.8549	**0.9238**	0.8405
SeaNet [43]	TGRS2023	**0.0073**	0.8649	0.8519	**0.8304**	**0.9710**	**0.9651**	**0.9602**	0.9208	———
ERPNet [21]	TCYB2023	0.0089	0.8897	0.8565	0.7788	0.9599	0.9399	0.9228	0.9201	**0.8514**
PROFILE	———	**0.0070**	**0.9053**	**0.8909**	**0.8646**	0.9669	**0.9627**	**0.9333**	**0.9305**	**0.8886**

2) EORSSD Dataset: For the EORSSD dataset, an extended version of the ORSSD database contains 2000 optical samples. Here, 70% of the images have been randomly sourced during training, while the remaining 30% have been designated for testing purposes.

B. Experimental Settings

At this work, our PROFILE is trained using an SGD optimizer, which is capable of updating the model parameters iteratively. During the training phase, we initialize the learning rate of the Swin Transformer to 0.003 and 0.03 for others. The learning rate initially increases and then decays during the training process, and the momentum and weight decay have been set to 0.9 and 5×10^{-4}. We set the batch size to 16, and the number of epochs is 32. Additionally, the dimensions of all training and testing images are uniformly adjusted to 448×448. For the experiments, all approaches have been trained and tested regarding the ORSSD and EORSSD datasets. PyTorch 1.12.1 is used in all experiments, which are powered by an Intel[1] Xeon[1] E5-2680 v4 CPU at 2.40 GHz operating under Ubuntu 18.04, with an Nvidia GeForce RTX 3090 GPU.

C. Evaluation Metrics

To rate the potency of our approach, ten wide metrics are selected in this experiment, that is, maximum F-measure ($F_{\max}$) [9], adaptive F-measure (F_{adp}) [9], maximum E-measure ($E_{\max}$) [51], mean absolute error (MAE) [52], mean E-measure (E_{mean}) [51], mean F-measure (F_{mean}) [9], adaptive E-measure (E_{adp}) [51], S-measure (S_m) [53], Fbw-measure (F_{bw}) [9], and *Precision–Recall curve* (PR curve) [9] are adopted to comprehensively measure the performance of our PROFILE and other competitors. Especially, better detection performance results from the PR curve being closer to the upper right, whereas for other metrics, the larger the $F_{\max}$, F_{mean}, F_{adp}, $E_{\max}$, E_{mean}, E_{adp}, S_m, and F_{bw} represent improved performance, and a smaller *MAE* value implying better performance. The evaluation code used in the experiments is based on Python (download link: https://github.com/zyjwuyan/SOD_Evaluation_Metrics).

D. Comparative Analysis of State-of-the-Art Methods

In this section, to evaluate the performance of the proposed PROFILE against other approaches, we conducted a comparative analysis of 20 classical SOD methods, including seven SOD approaches applied to optical RSIs (e.g., DAFNet [12], ERPNet [21], LVNet [10], EMFI [41], RRNet [54], SeaNet [43], and MJRBM [55]) and 13 SOD methods for NSIs (i.e., MINet [32], PGNet [17], DSS [56], PoolNet [57], GateNet [58], RADF [59], R3Net [60], EGNet [13], GCPA [61], ITSD [62], U2Net [63], PAKRN [64], and SUCA [65]). Among them, all algorithms are based on deep-learning methods. To ensure fairness in our comparison, we utilized the parameters and saliency maps provided by the respective authors.

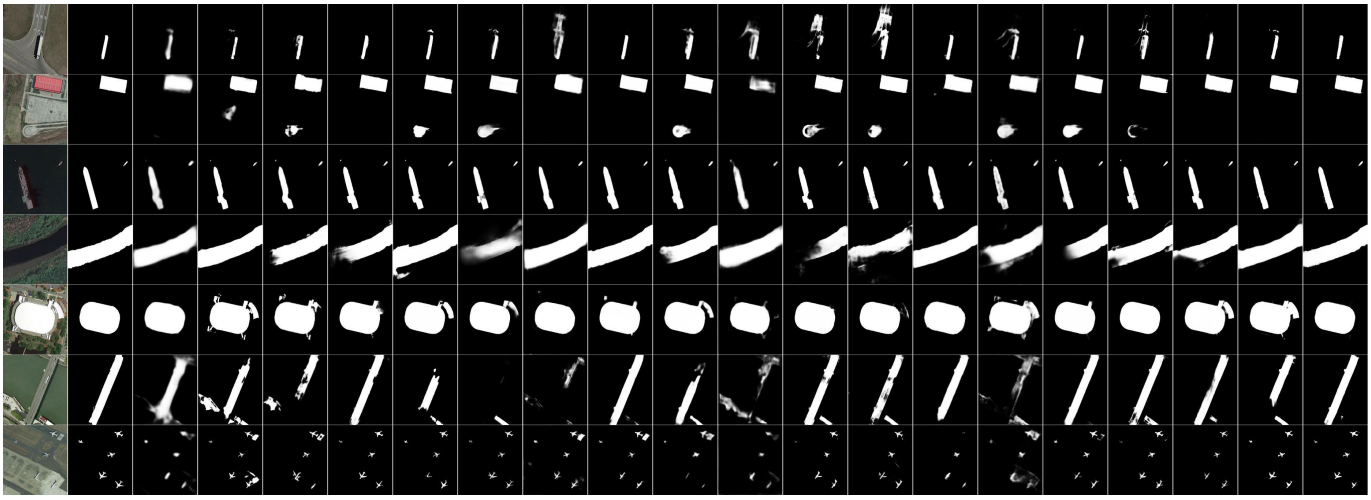
1) Quantitative Comparison: We have summarized the results of various saliency detectors on two benchmark datasets in terms of MAE, $F_{\max}$, F_{mean}, F_{adp}, $E_{\max}$, E_{mean}, E_{adp}, S_m, and F_{bw} in Tables I and II. Notably, the PROFILE achieves superior performance on almost all evaluation metrics among competitors. Particularly, the proposed PROFILE outperforms the ERPNet by 8.58%, 1.56%, and 2.28% on the EORSSD dataset in terms of F_{adp}, $F_{\max}$, E_{mean}, and S-measure, respectively. Additionally, several other saliency detectors, such as EGNet, PoolNet, EMFI, and ERPNet, use edge information in guiding salient object predictions. It is evident that our PROFILE also drastically outranks these competitors. Especially, compared to the latest method, ERPNet, as evidenced by Table I, on $F_{\max}$ (0.8897 versus 0.9053),

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AVERAGE ACCURACY COMPARISON ON THE ORSSD DATASET. THE FIRST THREE BEST PREDICTIONS HAVE BEEN RENDERED IN **RED**, **BLUE**, AND **BLACK**

Method	Publication	ORSSD Dataset								
		$MAE \downarrow$	$F_{max} \uparrow$	$F_{mean} \uparrow$	$F_{adp} \uparrow$	$E_{max} \uparrow$	$E_{mean} \uparrow$	$E_{adp} \uparrow$	$S_m \uparrow$	$F_{bw} \uparrow$
DSS [56]	CVPR2017	0.0363	0.7619	0.7115	0.6333	0.8860	0.8368	0.8085	0.8260	0.6657
R3Net [60]	IJCAI2018	0.0399	0.7608	0.7529	0.7529	0.8903	0.8674	0.8687	0.8143	0.7400
RADF [59]	AAAI2018	0.0382	0.7775	0.7010	0.5847	0.9130	0.8307	0.7678	0.8259	0.6307
PoolNet [57]	CVPR2019	0.0358	0.7825	0.7116	0.6292	0.9311	0.8621	0.8124	0.8394	0.6487
EGNet [13]	ICCV2019	0.0217	0.8468	0.7603	0.6584	0.9720	0.8959	0.8226	0.8725	0.7039
LVNet [10]	TGRS2019	0.0208	0.8432	0.8162	0.7659	0.9456	0.9259	0.9195	0.8813	0.7935
GCPA [61]	AAAI2020	0.0171	0.8856	0.8563	0.7974	0.9496	0.9307	0.9205	0.9013	0.8410
GateNet [58]	ECCV2020	0.0137	0.9052	0.8860	0.8397	0.9664	0.9539	0.9278	0.9193	0.8718
ITSD [62]	CVPR2020	0.0165	0.8913	0.8679	0.8232	0.9601	0.9483	0.9235	0.9043	0.8470
U2Net [63]	PR2020	0.0166	0.8917	0.8669	0.8202	0.9539	0.9386	0.9326	0.9156	0.8551
MINet [32]	CVPR2020	0.0146	0.8933	0.8715	0.8373	0.9531	0.9423	0.9273	0.9031	0.8601
DAFNet [12]	TIP2021	**0.0113**	0.9111	0.8690	0.8037	**0.9771**	0.9540	0.9160	0.9192	0.8524
PAKRN [64]	AAAI2021	0.0139	0.9072	0.8909	**0.8722**	0.9680	0.9620	0.9379	0.9231	0.8791
SUCA [65]	TMM2021	0.0145	0.8653	0.8398	0.7906	0.9583	0.9391	0.9094	0.8989	0.8172
PGNet [17]	CVPR2022	0.0225	0.8338	0.8192	0.7921	0.9420	0.9218	0.9105	0.8894	——
EMFI [41]	TGRS2022	0.0109	0.9185	**0.9038**	**0.8793**	0.9737	**0.9671**	**0.9663**	**0.9365**	**0.8950**
MJRBM [55]	TGRS2022	0.0163	0.9023	0.8749	0.8185	0.9623	0.9418	0.9328	0.9194	0.8602
RRNet [54]	TGRS2022	0.0117	**0.9191**	0.8875	0.8267	0.9721	0.9506	0.9204	**0.9330**	0.8767
SeaNet [43]	TGRS2023	**0.0105**	0.8942	0.8772	0.8625	**0.9767**	**0.9722**	**0.9670**	0.9260	——
ERPNet [21]	TCYB2023	0.0135	**0.9157**	**0.8926**	0.8526	0.9710	0.9566	0.9520	0.9247	**0.8843**
PROFILE	——	**0.0094**	**0.9324**	**0.9153**	**0.8956**	**0.9780**	**0.9731**	**0.9537**	**0.9389**	**0.9063**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m) (n) (o) (p) (q) (r) (s) (t) (u)

Fig. 4. Comparison of prediction results on the ORSSD dataset employing several frontier methods. (a) Input RSIs. (b) Saliency maps. (c) EGNet. (d) R3Net. (e) LVNet. (f) PGNet. (g) EMFI. (h) MJRBM. (i) DSS. (j) U2Net. (k) SUCA. (l) RADF. (m) MINet. (n) GateNet. (o) DAFNet. (p) PoolNet. (q) GCPA. (r) ITSD. (s) RRNet. (t) PAKRN. (u) PROFILE.

F_{mean} (0.8565 versus 0.8909), F_{adp} (0.7788 versus 0.8646), E_{max} (0.9599 versus 0.9669), E_{mean} (0.9399 versus 0.9627), E_{adp} (0.9228 versus 0.9333), S_m (0.9201 versus 0.9305), and F_{bw} (0.8514 versus 0.8886). At the same time, the MAE dropped from 0.0089 (ERPNet) to 0.0070 (PROFILE). Despite having inferior performance on E_{mean} and E_{adp} compared to SeaNet, and lower performance on MAE and E_{max} compared to DAFNet, we noticeably outscore those methods in all other metrics.

As illustrated in Table II, the proposed technique boasts substantial improvements against other competitors regarding the ORSSD dataset. It achieves the third-best E_{adp} score and attains the optimal experimental results in all other indicators. Especially, compared with other RSI-SOD approaches, that is, DSS, MINet, EMFI, MJRBM, and ERPNet, the proposed PROFILE is significantly better than them on F_{max}, E_{max}, S_m, and F_{bw} ([17.05%, 9.2%, 11.29%, 24.06%], [3.91%, 2.49%, 3.58%, 4.62%], [1.39%, 0.43%, 0.24%, 1.13%], [3.01%, 1.57%, 1.96%, 4.61%], and [1.67%, 0.7%, 1.42%, 2.2%], respectively). Additionally, we have gained an MAE score that is 0.11% less than the second-best model (SeaNet), the F_{mean} score is up 2.27% above the third-best model (ERPNet), and the E_{mean} score is incremented by 0.6% against the third-best model (EMFI). In a nutshell, Tables I and II have revealed that PROFILE exhibits the optimal properties across two datasets, yielding a SOTA benchmark for comparison with RSI-SOD.

2) Qualitative Comparison: To better showcase the novelty of the PROFILE, 20 frequently performed methods have been selected for testing on the ORSSD and EORSSD databases, and typical predictions can be viewed in Figs. 4 and 5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

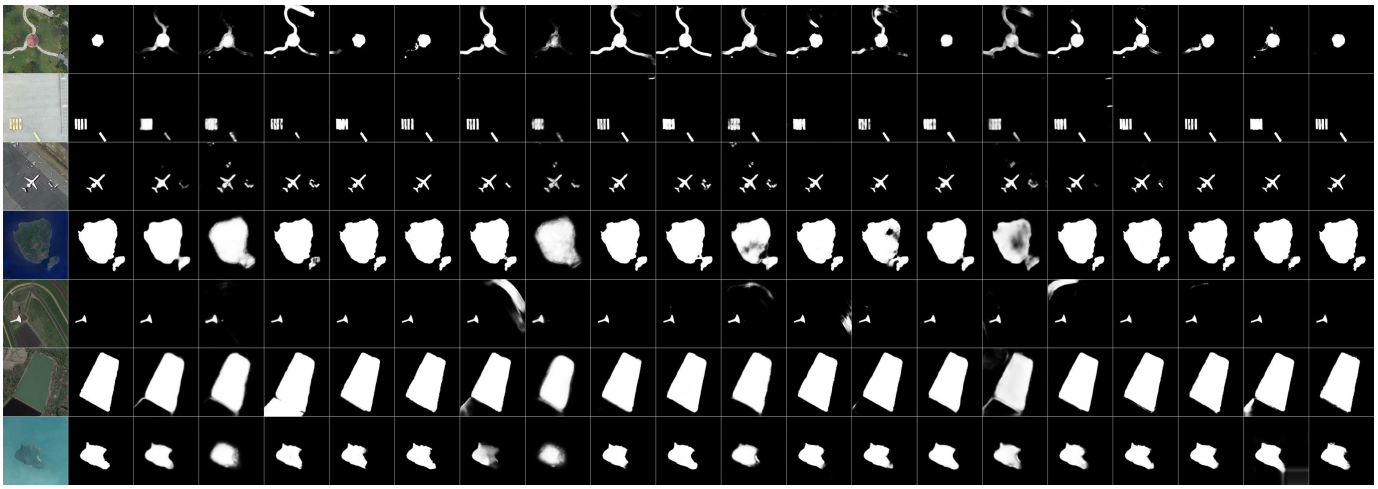


Fig. 5. Comparison of prediction results on the EORSSD dataset employing several frontier methods. (a) Input RSIs. (b) Saliency maps. (c) EGNet. (d) R3Net. (e) LVNet. (f) PGNet. (g) EMFI. (h) MJRBM. (i) DSS. (j) U2Net. (k) SUCA. (l) RADF. (m) MINet. (n) GateNet. (o) DAFNet. (p) PoolNet. (q) GCPA. (r) ITSD. (s) RRNet. (t) PAKRN. (u) PROFILE.

For these results, a variety of complex scenes are depicted, involving large-scale objects, smaller objects, multiobjects, objects with low contrast, and objects with intricate contours. Overall, as is clearly seen, our PROFILE consistently yields the best predictions that closely resemble the ground truth across various changing scenarios. More particularly, our proposed method efficiently detects salient objects more completely while retaining their sharp edges. The PROFILE has the following superiority and competitiveness.

a) Advantages in content integrity: It is notable that the trucks, roofs, rivers, and bridges on rows first and fourth of Fig. 4 contain structural information that can be difficult to understand, and the tributaries are relatively narrow. Several existing methods, such as DSS, RADF, MINet, GateNet, and PoolNet, have failed to accurately detect the salient regions. Conversely, the proposed PROFILE can precisely identify salient regions in the second and sixth rows of the image, which contain irregular protruding objects, while preserving the integrity of their edge boundary information. Compared to competitors, that is, R3Net, LVNet, MINet, GateNet, and PoolNet, which undergo the identification of the background as salient regions or fail to detect incomplete results. The results of the experiments reveal that the employed PROFILE consistently recognizes the salient region/objects, effectively minimizing background confusion. Empirical evidence demonstrates that MHCCF progressively integrates multitiered image features, augmenting the feature representation of pertinent regions, thus enabling the model's triumphant adaptation to demanding multiscale target scenarios. Moreover, it facilitates precise and comprehensive inferences for RSI-SOD tasks.

b) Advantages in challenging scenarios: The images in rows third, fifth, and seventh of Fig. 4 show the presence of multiple objects, with a noticeable effect of low contrast, particularly in rows third and seventh. The method employed has proven effective in accurately identifying and locating saliency areas characterized by complex boundaries or anomalous topography in the midst of complex backgrounds or low-contrast situations. For example, in the third image, R3Net, EMFI, MJRBM, MINet, DAFNet, and RRNet cannot detect object areas precisely and suffer from issues, that is, edge distortion and object positioning errors. Similarly, in Fig. 4, for the 2nd and 4th samples, the EGNet, R3Net, PGNet, and PAKRN either incorrectly accentuate background regions or fail to clearly distinguish salient objects. Especially, DSS, SUCA, MINet, and PoolNet generally perform incomplete detection or identify salient regions that are characterized by blurred texture and edge structure. Evidently, owing to the inherent advantage of implicit supervision by BCCM over the target region and its boundaries, PROFILE can achieve the comprehensiveness of salient object information in complex scenarios.

c) Advantages in location accuracy: In the 1st, 3rd, 5th, 6th, and 7th images of Fig. 5, several representative works [e.g., EGNet, R3Net, DSS, and PoolNet shown in Fig. 5(c), (d), (i), and (p)] fail to accurately identify small salient objects in the images, whereas the other methods [e.g., LVNet, MJRBM, U2Net, SUCA, RADF, and GateNet shown in Fig. 5(e), (h), (j)–(l), and (n)] either incorrectly accentuate background regions or fail to clearly distinguish salient objects. The results depicted in Fig. 5 demonstrate that our approach effectively successfully identifies and localizes small saliency objects (SSOs) while mitigating background interference. The above-mentioned experimental results reveal that IACTer can effectively capture spatial intricacies and semantic relationships, making it applicable for detecting intricate backgrounds and low-contrast environments.

3) PR and F-Measure Curves: Apart from aforementioned nine quantitative metrics (MAE, $F_{\max}$, F_{mean}, F_{adp}, $E_{\max}$, E_{mean}, E_{adp}, S_m, and F_{bw}), the comparative analysis of PR curves and F-measure curves for various techniques on the two publicly available SOD benchmark databases are also presented in Figs. 6–9. The visualization presented in Figs. 6 and 7 indicates that high levels of precision and recall values are exhibited by our PROFILE, as well as minimal discrepancies among the two assessment metrics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

TABLE III

ANALYSIS OF THE ABLATION STUDY BASED ON THE EORSSD DATABASE IN QUANTITATIVE TERMS. THE FIRST THREE BEST PREDICTIONS HAVE BEEN RENDERED IN **RED**, **BLUE**, AND **BLACK**

Ablation	Configurations	$MAE \downarrow$	$F_{max} \uparrow$	$F_{mean} \uparrow$	$F_{adp} \uparrow$	$E_{max} \uparrow$	$E_{mean} \uparrow$	$E_{adp} \uparrow$	$S_m \uparrow$
Architecture	PGNet	0.0153	0.7887	0.7775	0.7402	0.9315	0.9178	0.8858	0.8757
	IACTer	0.0131	0.8408	0.7994	0.7316	0.9450	0.9271	0.8617	0.8843
	+MHCCF	0.0070	0.8994	0.8822	**0.8544**	0.9660	0.9612	0.9277	0.9266
	PROFILE	0.0070	**0.9053**	**0.8909**	**0.8646**	**0.9669**	**0.9627**	**0.9333**	**0.9305**
$loss$	$+ \ell_p \ loss$	**0.0066**	**0.9030**	0.8858	0.8510	**0.9679**	**0.9624**	0.9251	**0.9307**
	$+ \ell_s \ loss$	**0.0064**	**0.9020**	**0.8871**	0.8541	0.9684	0.9617	**0.9302**	**0.9309**
	$+ \ell_{mse} \ loss$	**0.0069**	0.9015	**0.8860**	**0.8546**	**0.9668**	0.9615	**0.9286**	0.9295

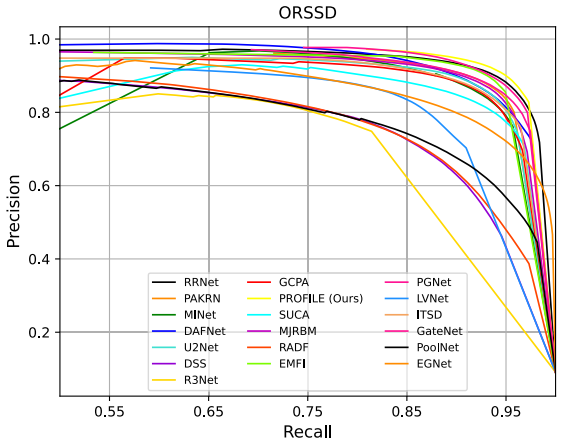


Fig. 6. Analysis of the PR curves obtained by applying the PROFILE and 18 SOD methods to the ORSSD.

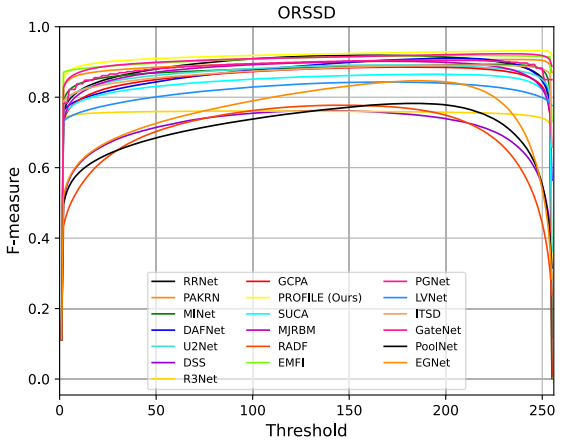


Fig. 8. Analysis of the F_{measure} curves obtained by applying the PROFILE and 18 SOD methods to the ORSSD.

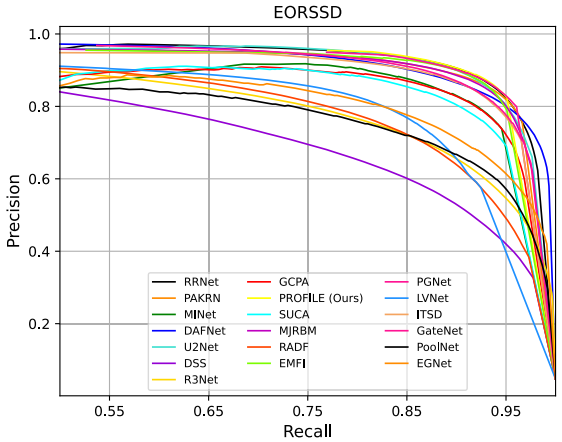


Fig. 7. Analysis of the PR curves obtained by applying the PROFILE and 18 SOD methods to the EORSSD.

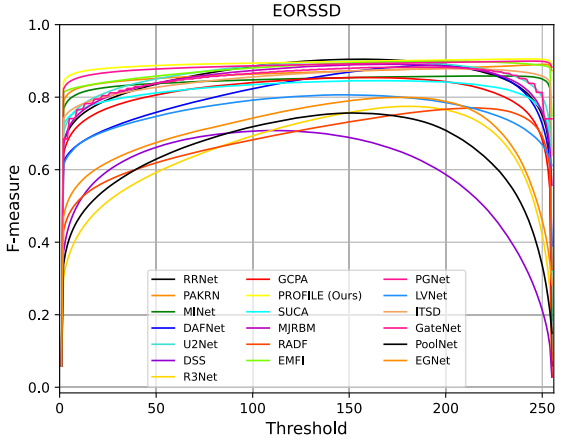


Fig. 9. Analysis of the F_{measure} curves obtained by applying the PROFILE and 18 SOD methods to the EORSSD.

The results of the proposed PROFILE on both databases are in proximity to the coordinate (1, 1), which indicates the fact that our model exhibits superior performance compared to other competing methods. Furthermore, as depicted in Figs. 8 and 9, our PROFILE's F-measure curve encompasses the most substantial coordinate region compared to all other evaluated approaches, thereby unequivocally showcasing the efficacy of our PROFILE.

E. Ablation Study

For authenticating the efficacy of each constituent element, an ablation experiment is performed on the EORSSD dataset by omitting the various modules within the proposed PROFILE. Additionally, BCCM also discussed the impact of various loss functions with respect to the model properties. Among the three best results listed in Table III, the top two are highlighted in **red**, **blue**, and the third in **bold**.

1) Validation of IACTer: The "Baseline" model is the IACTer used for comparison without any gimmicks. The notation "+MHCCF" stands for the "Baseline" model with MHCCF, while "+BCCM" indicates the "Baseline" model with MHCCF and BCCM. Notwithstanding the poor performance of the baseline model on all metrics, when compared to other approaches, that is, PGNet, it is nevertheless a reasonably robust approach. From Table III, we can see that the MAE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

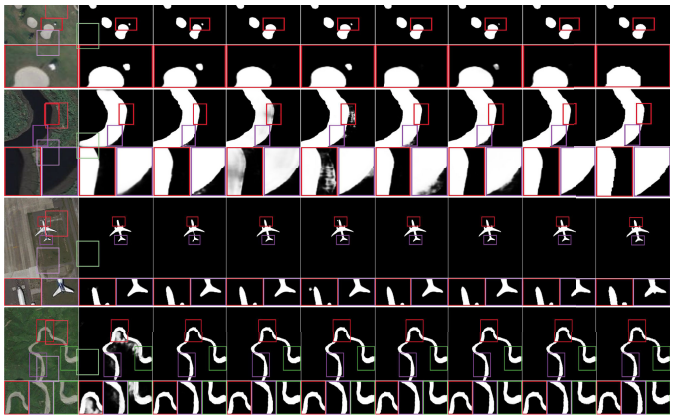


Fig. 10. Qualitative comparison of the proposed PROFILE with ablation studies, rows 1, 3, 5, and 7 show the results of different modules, while rows 2, 4, 6, and 8 depict zoomed images of the red rectangles. (a) Optical RSIs. (b) PGNet. (c) IACTer. (d) IACTer + MHCCF. (e) IACTer + MHCCF + ℓ_p. (f) IACTer + MHCCF + ℓ_p + ℓ_s. (g) IACTer + MHCCF + ℓ_p + ℓ_s + ℓ_{mse}. (h) PROFILE. (i) Ground truth.

score has decreased from 0.0153 to 0.0131, the F_{max} value has improved from 0.7887 to 0.8408, and the F_{mean} score has increased from 0.7775 to 0.7994. As the results of the visualization experiment are shown in Fig. 10(b) and (c), the results of PGNet have emerged, identifying background regions as significant targets or failing to detect complete results. In particular, the last row exhibits a significant discrepancy in the localization of the salient objects within the mountain images, with the saliency regions displaying indistinct boundaries. Nevertheless, the base method can avoid this problem well, and it also proves the effectiveness of our IACTer.

2) Validation of MHCCF: By comparing *row* 2 with *row* 3 in Table III, it is evident that the incorporation of the MHCCF module results in an improvement of 0.61% in terms of MAE for the RSI-SOD task. Moreover, we observe a ascend in performance [F_{max}: 0.8408 versus 0.8994], [F_{mean}: 0.7994 versus 0.8822], [F_{adp}: 0.7316 versus 0.8544], [E_{max}: 0.9450 versus 0.9660], [E_{mean}: 0.9271 versus 0.9612], [E_{adp}: 0.8617 versus s 0.9277], and [S_m: 0.8843 versus 0.9266]. Based on these results, the validity of the proposed MHCCF was verified. The comparison of the "baseline" model and the +MHCCF visual prediction results that are illustrated in Fig. 10(c) and (d). Clearly, the issue of inaccurate protruding region localization in the base model can be alleviated by +MHCCF. Nevertheless, the potential for improvement remains in the completeness of salient region/object identification.

3) Validation of BCCM: To ascertain the soundness of our BCCM module, we commence by outlining three distinct variations. These include: 1) our model that is furnished solely with ℓ_p ($\ell_{\text{BCE}}+\ell_{\text{IoU}}+\ell_{\text{ag}}+\ell_p$); 2) our model that is equipped exclusively with ℓ_s ($\ell_{\text{BCE}}+\ell_{\text{IoU}}+\ell_{\text{ag}}+\ell_p+\ell_s$); and 3) our model that solely employs with ℓ_{MSE} ($\ell_{\text{BCE}}+\ell_{\text{IoU}}+\ell_{\text{ag}}+\ell_p+\ell_s+\ell_{\text{MSE}}$). The results presented in Table III reveal that the optimal detection properties have been gained by our PROFILE with +BCCM [No. 8 in Fig. 10]. When employing ℓ_{MSE}, our method achieves the smallest MAE value and the largest S_m value than $+\ell_p$ and $+\ell_s$. Besides, we also verify the desirability of the proposed various optimization functions and the impact of the BCCM on the performance of the proposed PROFILE. Similarly, in Fig. 10, it also demonstrates that our method can perform accurate localization and complete identification of the location of salient objects under complex background and imaging conditions (e.g., shooting time and illumination intensity), and the edge information of the salient region is more consistent with ground truth, for example, two engines of each airplane [No. 5 in Fig. 10] and mountain range [No. 7 in Fig. 10]. Furthermore, it also meets our motivations to a certain extent.

TABLE IV

COMPARATIVE STUDY OF OUR METHOD WITH THE MOST ADVANCED METHODS WITH REGARD TO CODE, FLOPS, PARAMS, TIME, AND F_{MAX} ON THE ORSSD DATABASE. IT SHOULD BE NOTED THAT "P" STANDS FOR PYTORCH, AND "T" INDICATES TENSORFLOW

Methods	Code	FLOPs (G)	Params(M)	Time	F_{max}
PoolNet [57]	P	115.09	68.56	**0.01**	0.7825
EGNet [13]	P	287.67	111.66	0.03	0.8468
GCPA [61]	P	54.31	67.06	**0.01**	0.8856
GateNet [58]	P	162.07	128.63	0.02	0.9052
MINet [32]	P	87.03	162.38	0.02	0.8933
DAFNet [12]	P	839.21	**29.35**	0.04	0.9111
SUCA [65]	P	88.57	115.58	**0.01**	0.8653
LVNet [10]	T	——-	207	0.74	0.8432
DSS [56]	P	——-	248	0.12	0.7619
U2Net [63]	P	——-	168	0.04	0.8917
R3Net [60]	P	——-	142	0.48	0.7608
RADF [59]	P	——-	248	0.15	0.7775
EMFI [41]	P	541.05	95.09	0.04	0.9185
MJRBM [55]	P	155.09	43.78	0.03	0.9302
RRNet [54]	P	692.15	86.27	0.02	0.9191
ACNet [66]	P	122.26	57.98	——-	0.9238
PROFILE	P	**19.34**	89.75	**0.01**	**0.9324**

F. Model Complexity and Efficiency Analysis

Benchmarks our network against other in-vogue models as regards floating point operations (FLOPs), parameters (Params), and interference time. Here, all experiments are undertaken on an NVIDIA GeForce RTX 3090 GPU. From Table IV, we can see that the computed parameters of the PROFILE are 89.75 M, which is preferable to EGNet, MJRBM, GateNet, MINet, SUCA, LVNET, DSS, RADF, and EMFI. Additionally, our model shows faster detection speed, and its FLOPS are inferior to PGNet, albeit it is nonetheless the appropriate model to be applied in real-time. A noteworthy feature of our approach is that, despite having more parameters and larger FLOPS values than PGNET, exhibits a 9.89% improvement in F_{max}. Overall, as the results listed in Table IV, the proposed PROFILE is highly efficient and profitable, boasting significantly faster inference speeds, fewer parameters, and lower computational costs than other methods. However, the presented approach's considerable improvement potential exists from an alternative standpoint. Specifically, it needs to improve detection accuracy while minimizing the computational parameters of the model.

G. Discussion

1) Failure Cases: As previously stated, the proposed PROFILE displays commendable ability in RSI-SOD.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

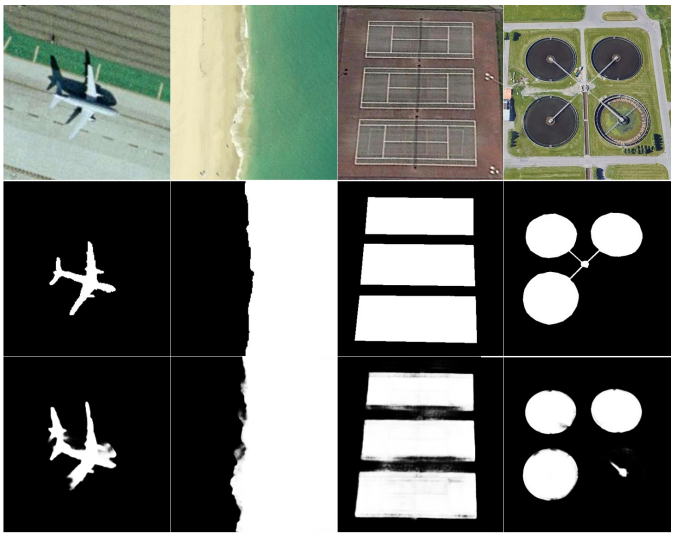


Fig. 11. Failure case predictions from the EORSSD database. The top row shows the input RSIs, the following row gives the real saliency maps, and its final row shows the PROFILE saliency maps.

Nonetheless, in several instances of particularly challenging scenes, our method falls short of providing flawless predictions, as evidenced by the sample images in Fig. 11. In the 1st column, for instance, on the fuselage of the aircraft, shadows are suppressed; however, on the rear, they are mistakenly preserved. As shown in columns 2 and 3, our PROFILE is capable of detecting the general structure of some salient objects. However, the edges of the river and badminton court are blurred, owing to the similarity in appearance of the two targets and their surroundings. In the fourth column of Fig. 11, the storage tanks and transport tubes are enveloped by a cluttered background, where the two objects are also visually akin to the surroundings. Three scenarios can be used to categorize these failure cases: 1) our method encounters challenges when dealing with complex shadows; 2) detecting salient objects with indistinct or blurred edges is still problematic; and 3) it is nevertheless difficult to discern a salient object/region when both the object itself and its surrounding exhibit almost identical appearances.

2) Future Work: In the coming times, we contend that the aforementioned issue can be mitigated by implementing the following potential solutions. One effective approach would involve the creation of a benchmark dataset for RSI-SOD, which includes a wider range of images with cluttered and similar backgrounds when compared to the EORSSD dataset. This is a straightforward solution because the performance of the supervised-based approaches is positively correlated with the quality of the dataset. Hence, a massive amounts dataset comprising many different types of images closely resembles real-world situations and could aid in reducing failure cases. Additionally, one may contemplate the introduction of unsupervised learning [67] to effectively mitigate the reliance on labeled data, addressing the issue of dependency on annotated datasets. Another solution could be to provide a structural supplement by introducing auxiliary information containing geometric and spatial details, that is, depth maps [68], and edge information [69]. These aids can assist the network in identifying the interrelations of different positions and distinguishing among salient or nonsalient items, thereby mitigating the issues described earlier. Furthermore, in the future, for enhanced performance assessment of the RSI-SOD task, we contemplate the incorporation of evaluation metrics such as the 3-D ROC curve [70] within the RSI-SOD task.

V. Conclusion

This article introduces a novel PROFILE for RSI-SOD, incorporating three key elements: IACTer, MHCCF, and BCCM. Through rigorous experimentation, it is demonstrated that the proposed PROFILE surpasses state-of-the-art SOD approaches, addressing the challenges of accurate positioning and comprehensive detection of salient objects within complex backgrounds and imaging environments to a considerable extent. However, there is still room for improvement, particularly in cases where the background of salient objects is exceptionally intricate. Future advancements could involve incorporating geometric and spatial details or constructing datasets with more cluttered backgrounds and varying illumination conditions. Furthermore, this undertaking could also serve the compression process of critical regions in satellite imagery, thereby enhancing the efficiency of remote-sensing data transmission and reducing the amount of data transmitted.

References

[1] X. Li, D. Song, and Y. Dong, "Hierarchical feature fusion network for salient object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 9165–9175, 2020.

[2] X. Li, G. Li, and B. Zhao, "Low-light hyperspectral image enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537013.

[3] B. Sun, G. Liu, and Y. Yuan, "F3-Net: Multiview scene matching for drone-based geo-localization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 56106113.

[4] G. Liu, Y. Yuan, Y. Zhang, Y. Dong, and X. Li, "Style transformation-based spatial–spectral feature learning for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5401515.

[5] Y. Wu, K. Zhang, J. Wang, Y. Wang, Q. Wang, and X. Li, "GCWNet: A global context-weaving network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619912.

[6] B. Wang, Y. Zhao, and X. Li, "Multiple instance graph learning for weakly supervised remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613112.

[7] M.-M. Cheng, N. J. Mitra, X. Huang, P. H. S. Torr, and S.-M. Hu, "Global contrast based salient region detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 569–582, Mar. 2015.

[8] J. Han, D. Zhang, X. Hu, L. Guo, J. Ren, and F. Wu, "Background prior-based salient object detection via deep reconstruction residual," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 8, pp. 1309–1321, Aug. 2015.

[9] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.

[10] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.

[11] Z. Huang, H. Chen, B. Liu, and Z. Wang, "Semantic-guided attention refinement network for salient object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 11, p. 2163, May 2021.

[12] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

[13] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8778–8787.

[14] P. Han, B. Zhao, and X. Li, "Edge-guided remote-sensing image compression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5524515.

[15] S. Tang, J. Zhang, S. Zhu, and P. Tan, "QuadTree attention for vision transformers," 2022, *arXiv:2201.02767*.

[16] Y. Wang, Y. Qiu, P. Cheng, and J. Zhang, "Hybrid CNN-transformer features for visual place recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 1109–1122, Mar. 2023.

[17] C. Xie, C. Xia, M. Ma, Z. Zhao, X. Chen, and J. Li, "Pyramid grafting network for one-stage high resolution saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11707–11716.

[18] M. H. Sharif, D. Demidov, A. Hanif, M. Yaqub, and M. Xu, "TransResNet: Integrating the strengths of ViTs and CNNs for high resolution medical image segmentation via feature grafting," in *Proc. Brit. Mach. Vis. Conf., 2022*, p. 293.

[19] B. Zhao, P. Han, and X. Li, "Vehicle perception from satellite," *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1–10, Nov. 2023, doi: 10.1109/TPAMI.2023.3335953.

[20] Z. Liu, D. Zhao, Z. Shi, and Z. Jiang, "Unsupervised saliency model with color Markov chain for oil tank detection," *Remote Sens.*, vol. 11, no. 9, p. 1089, May 2019.

[21] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.

[22] N. Liu and J. Han, "DHSNet: Deep hierarchical saliency network for salient object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 678–686.

[23] W. Zhu, S. Liang, Y. Wei, and J. Sun, "Saliency optimization from robust background detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2814–2821.

[24] D. A. Klein and S. Frintrop, "Center-surround divergence of feature statistics for salient object detection," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2214–2219.

[25] T. Liu, J. Sun, N.-N. Zheng, X. Tang, and H.-Y. Shum, "Learning to detect a salient object," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2007, pp. 1–8.

[26] D. Zhao, J. Wang, J. Shi, and Z. Jiang, "Sparsity-guided saliency detection for remote sensing images," *J. Appl. Remote Sens.*, vol. 9, no. 1, Sep. 2015, Art. no. 095055.

[27] G. Zhu, Q. Wang, Y. Yuan, and P. Yan, "Learning saliency by MRF and differential threshold," *IEEE Trans. Cybern.*, vol. 43, no. 6, pp. 2032–2043, Dec. 2013.

[28] H. Peng, B. Li, H. Ling, W. Hu, W. Xiong, and S. J. Maybank, "Salient object detection via structured matrix decomposition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 818–832, Apr. 2017.

[29] L. Zhang and K. Yang, "Region-of-interest extraction based on frequency domain analysis and salient region detection for remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 916–920, May 2014.

[30] L. Ma, B. Du, H. Chen, and N. Q. Soomro, "Region-of-interest detection via superpixel-to-pixel saliency analysis for remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1752–1756, Dec. 2016.

[31] L. Zhang, A. Li, Z. Zhang, and K. Yang, "Global and local saliency analysis for the extraction of residential areas in high-spatial-resolution remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 3750–3763, Jul. 2016.

[32] Y. Pang, X. Zhao, L. Zhang, and H. Lu, "Multi-scale interactive network for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9410–9419.

[33] Y. Liu, P. Wang, Y. Cao, Z. Liang, and R. W. H. Lau, "Weakly-supervised salient object detection with saliency bounding boxes," *IEEE Trans. Image Process.*, vol. 30, pp. 4423–4435, 2021.

[34] H. Zhou, P. Chen, L. Yang, X. Xie, and J. Lai, "Activation to saliency: Forming high-quality labels for unsupervised salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 2, pp. 743–755, Feb. 2023.

[35] R. Yasarla, R. Weng, W. Choi, V. Patel, and A. Sadeghian, "3SD: Self-supervised saliency detection with no labels," 2022, *arXiv:2203.04478*.

[36] Y. Wang, W. Zhang, L. Wang, T. Liu, and H. Lu, "Multi-source uncertainty mining for deep unsupervised saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11717–11726.

[37] S. Li, F. Liu, L. Jiao, X. Liu, and P. Chen, "Learning salient feature for salient object detection without labels," *IEEE Trans. Cybern.*, vol. 53, no. 2, pp. 1012–1025, Feb. 2023.

[38] X. Qin, Z. Zhang, C. Huang, C. Gao, M. Dehghan, and M. Jagersand, "BASNet: Boundary-aware salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7471–7481.

[39] J. Li, S. Qiao, Z. Zhao, C. Xie, X. Chen, and C. Xia, "Rethinking lightweight salient object detection via network depth-width tradeoff," 2023, *arXiv:2301.06679*.

[40] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[41] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605315.

[42] R. Xu, C. Wang, J. Zhang, S. Xu, W. Meng, and X. Zhang, "RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 1052–1064, 2023.

[43] L. Gao, B. Liu, P. Fu, and M. Xu, "Adaptive spatial tokenization transformer for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602915.

[44] D. Harman, "Some experiments in color television," *Proc. Inst. Radio Eng.*, vol. 29, no. 7, pp. 457–466, 1941.

[45] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[46] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 5508614.

[47] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3241097.

[48] J. Ying, J. Liu, J. Chen, S. Cao, M. Hou, and Y. Chen, "Multi-scale fusion network: A new deep learning structure for elliptic interface problems," *Appl. Math. Model.*, vol. 114, pp. 252–269, Feb. 2023.

[49] T. Zeng, F. Luo, T. Guo, X. Gong, J. Xue, and H. Li, "Multilevel context feature fusion for semantic segmentation of ALS point cloud," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[50] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, "GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 794–803.

[51] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," 2018, *arXiv:1805.10421*.

[52] F. Perazzi, P. Krähenbühl, Y. Pritch, and A. Hornung, "Saliency filters: Contrast based filtering for salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 733–740.

[53] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4548–4557.

[54] R. Cong, Y. Zhang, L. Fang, J. Li, Y. Zhao, and S. Kwong, "RRNet: Relational reasoning network with parallel multiscale attention for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613311.

[55] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607913.

[56] Q. Hou, M.-M. Cheng, X. Hu, A. Borji, Z. Tu, and P. Torr, "Deeply supervised salient object detection with short connections," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5300–5309.

[57] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3912–3921.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.

[58] X. Zhao, Y. Pang, L. Zhang, H. Lu, and L. Zhang, "Suppress and balance: A simple gated network for salient object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 35–51.

[59] X. Hu, L. Zhu, J. Qin, C.-W. Fu, and P.-A. Heng, "Recurrently aggregating deep features for salient object detection," in *Proc. AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.

[60] Z. Deng et al., "R^3Net: Recurrent residual refinement network for saliency detection," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 684–690.

[61] Z. Chen, Q. Xu, and R. Cong, "Global context-aware progressive aggregation network for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 7, 2020, pp. 10599–10606.

[62] H. Zhou, X. Xie, J.-H. Lai, Z. Chen, and L. Yang, "Interactive two-stream decoder for accurate and fast saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9138–9147.

[63] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O. R. Zaiane, and M. Jagersand, "U2-Net: Going deeper with nested U-structure for salient object detection," *Pattern Recognit.*, vol. 106, Oct. 2020, Art. no. 107404.

[64] B. Xu, H. Liang, R. Liang, and P. Chen, "Locate globally, segment locally: A progressive architecture with knowledge review network for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 4, 2021, pp. 3004–3012.

[65] J. Li, Z. Pan, Q. Liu, and Z. Wang, "Stacked U-shape network with channel-wise attention for salient object detection," *IEEE Trans. Multimedia*, vol. 23, pp. 1397–1409, 2021.

[66] D. Song, Y. Dong, and X. Li, "Adjacent complementary network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607913.

[67] C.-I. Chang, Y.-M. Kuo, and P. F. Hu, "Unsupervised rate distortion function-based band subset selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023, Art. no. 5517218.

[68] L. Nie, C. Lin, K. Liao, S. Liu, and Y. Zhao, "Depth-aware multi-grid deep homography estimation with contextual correlation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 7, pp. 4460–4472, Jul. 2022.

[69] C. Yu, J. Huang, M. Song, Y. Wang, and C.-I. Chang, "Edge-inferring graph neural network with dynamic task-guided self-diagnosis for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3196311.

[70] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.



Pengfei Han is currently pursuing the Ph.D. degree with the School of Cybersecurity and School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an, China.

His research interests include remote-sensing imagery compression and data transmission.



Bin Zhao is an Associate Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China.

His research interests include introducing physics models and cognitive science to artificial intelligence.

Xuelong Li (Fellow, IEEE) is a Full Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:42:33 UTC from IEEE Xplore. Restrictions apply.